CHARAN TEJA VALAVALA

SOFTWARE, DATA AND APPLIED AI

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GitHub | Portfolio | LinkedIn

PROFILE

BITS CSE undergraduate building software and applied AI projects in document search, financial data, and incident workflows. Also completed machine-learning and reinforcement-learning coursework and contributed to an education platform used by 200+ students and trainees.

PROJECTS

Course Copilot — Python | FastAPI | MiniLM | FTS5 | Demo

- Built a document assistant using MiniLM dense embeddings, SQLite FTS5 BM25, reciprocal-rank fusion, PDF/TXT ingestion, and page-level evidence; optional Ollama generation checks citation references.
- On 52 manually labeled answerable queries across 13 notes, 2:1 BM25-to-dense fusion reached Hit@1 0.923, Hit@5 1.000, and MRR 0.962; 14/15 out-of-scope prompts abstained correctly.

FormScope — Python | FastAPI | SEC EDGAR | XBRL | Demo

- Built a source-linked SEC filings explorer that extracts XBRL company facts, validates annual filing periods, merges fallback concepts, computes revenue growth and net margin, and ranks 10-K passages.
- Checked annual revenue and net income against inline XBRL across 20 issuers; all 40 selected filing facts matched FormScope output.

IncidentPilot — Python | FastAPI | SQLite | Ollama | Demo

- Built a stateful incident-triage simulator with read-only log, metric, and runbook tools, evidence-linked diagnoses, versioned SQLite traces, and approval-gated simulated actions.
- Added a 20-case synthetic Ollama evaluation harness for tool-call and evidence-reference validity, diagnosis behavior, and fallback; no model performance score is claimed.

Diabetic Retinopathy Detection — Python | OpenCV | scikit-learn

- Built SVM classifiers for retinal fundus images from color and texture features; evaluated accuracy, precision, recall, and sensitivity.

Deep Reinforcement Learning for Object Tracking — Monte Carlo control | stochastic grid

- Implemented a stochastic grid environment with thermal states and teleportation; compared on-policy and off-policy Monte Carlo control and analyzed convergence.

EXPERIENCE AND EARLY-STAGE WORK

DataForge | Founder | concept stage — Feb-May 2026

- Developed an early AI infrastructure and data center venture concept with mentor guidance and participated in initial funding discussions.

MDDTI SBC | Web Development Intern — May-July 2024

- Contributed HTML/CSS frontend work to the organization website and online test platform, with MongoDB-backed features used by more than 200 students and trainees.

EDUCATION AND ACHIEVEMENTS

BITS Pilani, Hyderabad Campus | B.E. Computer Science Engineering — Expected 2027+
AP EAMCET | Statewide rank 8
Narayana Junior College | Class XII | 98.1% — 2022
Narayana High School | Class X | 10/10 GPA — 2020

TECHNICAL SKILLS

Programming: Python, C, C++, Java, JavaScript, SQL, HTML, CSS
Data and ML: NumPy, Pandas, Matplotlib, scikit-learn, OpenCV, Sentence Transformers, BM25/RRF
Backend and data: FastAPI, REST APIs, MongoDB, SQLite, SEC EDGAR/XBRL
Tools: Git, Docker, Ollama

RELEVANT COURSEWORK AND LANGUAGES

Artificial Intelligence, Machine Learning, Reinforcement Learning, Foundations of Data Science, Data Structures and Algorithms, Database Systems, Object-Oriented Programming. Spoken: English, Hindi, Telugu.